BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 128 · Programming: Arrays and Iteration Practice

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What makes a system embedded?

Q2. How many input combinations do three Boolean inputs have?
A. 3
B. 6
C. 9
D. 8

Q3. “Invalid test data must always have the wrong data type.” Correct this. Explain why.

Q4. Draw a structure diagram: Quiz contains Menu, Play and Results. Play contains Input and Scoring. Why is this not a flowchart of running steps?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Array iteration — A loop visits elements by their valid indexes.

Accumulator — A running total starts at zero and adds each element.

Counting condition — A counter increments only when a tested item matches a rule.

Off-by-one error — Accessing index length or missing final element changes the result or causes a fault.

New learning

Array iteration: A loop visits elements by their valid indexes.

Accumulator: A running total starts at zero and adds each element.

Counting condition: A counter increments only when a tested item matches a rule.

Off-by-one error: Accessing index length or missing final element changes the result or causes a fault.

Worked example

Calculate the total of [3,5,2,4] using a running total.

1. Start total=0.

2. After values 3 and 5, total=8.

3. Add 2 to reach 10, then 4 to reach 14.

4. Print 14; the four valid indexes are 0–3.

We Do · Practise with guidance

Guided 1

Count values greater than 5 in [2,7,9,4].

Worked method: Final count=2.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write the missing statement that adds the current file size.

```
array filesMB = [3,5,2]
total = 0
for i = 0 to 2
    ...
next i
print(total)
```

write

Question 1

Give the output of summing [4,1,3]. [2 marks]

Question 2

Explain why index 4 is invalid for a four-element zero-indexed array. [2 marks]

Question 3

Explain how to count even values in an array. [3 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A three-element zero-indexed array has last index 3.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It is built into a larger device for a specific task, often with limited resources.

Q2. D — 8

Q3. 12 is an integer but outside the valid range, so it must be rejected.

Q4. Quiz branches into Menu, Play and Results. Play branches into Input and Scoring. This shows parts and their hierarchy, rather than the order they run.

I Do / We Do pages · Guided practice

Q1. Final count=2.

You Do pages · Code First

```
array filesMB = [3,5,2]
total = 0
for i = 0 to 2
    total = total + filesMB[i]
next i
print(total)
```

Expected result: 10

You Do pages · Independent questions

Q1. 8.

Q2. Valid indexes are 0,1,2,3; index 4 is out of bounds.

Q3. Iterate each element; if value MOD 2 = 0 then increment counter; output counter.

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed: